

Finding 3: LLM agents’ trust behaviors have demographic biases on gender and races, demonstrate a relative preference for human over other LLM agents, are easier to undermine than to enhance, and may be influenced by reasoning strategies.

6 Implications

Implications for Human Simulation Human simulation is a strong tool in various applications of social science (Manning et al., 2024) and role-playing (Shanahan et al., 2023; Chen et al., 2024). Although plenty of works have adopted LLM agents to simulate human behaviors and interactions (Zhou et al., 2023; Gao et al., 2023b; Xu et al., 2024), it is still not clear enough whether LLM agents behave like humans in simulation. Our discovery of behavioral alignment between agent and human trust, which is especially high for GPT-4, provides important empirical evidence to validate the hypothesis that humans’ trust behavior, one of the most elemental and critical behaviors in human interaction across society, can effectively be simulated by LLM agents. Our discovery also lays the foundation for human simulations ranging from individual-level interactions to society-level social networks and institutions, where trust plays an essential role. We envision that behavioral alignment will be discovered in more kinds of behaviors beyond trust, and new methods will be developed to enhance behavioral alignment for better human simulation with LLM agents.

Implications for Agent Cooperation Many recent works have explored a variety of cooperation mechanisms of LLM agents for tasks such as code generation and mathematical reasoning (Li et al., 2023a; Zhang et al., 2023b; Liu et al., 2023). Nevertheless, the role of trust in LLM agent cooperation remains still unknown. Considering how trust has long been recognized as a vital component for cooperation in Multi-Agent Systems (MAS) (Ramchurn et al., 2004; Burnett et al., 2011) and across human society (Jones & George, 1998; Kim et al., 2022; Henrich & Muthukrishna, 2021), we envision that agent trust can also play an important role in facilitating the effective cooperation of LLM agents. In our study, we have provided ample insights regarding the intrinsic properties of agent trust, which can potentially inspire the design of trust-dependent cooperation mechanisms and enable the collective decision-making and problem-solving of LLM agents.

Implications for Human-Agent Collaboration Sufficient research has shown the advantage of human-agent collaboration in enabling human-centered collaborative decision-making (Cila, 2022; Gao et al., 2023c; McKee et al., 2022). Mutual trust between LLM agents and humans is important for effective human-agent collaboration. Although previous works have begun to study human trust towards LLM agents (Qian & Wexler, 2024), the trust of LLM agents towards humans, which could recursively impact human trust, is under-explored. In our study, we shed light on the nuanced preference of agents to trust humans compared with other LLM agents, which can illustrate the benefits of promoting collaboration between humans and LLM agents. In addition, our study has revealed demographic biases of agent trust towards specific genders and races, reflecting potential risks involved in collaborating with LLM agents.

Implications for the Safety of LLM Agents It has been acknowledged that LLMs achieve human-level performance in a variety of tasks that require high-level cognitive capacities such as memorization, abstraction, comprehension and reasoning, which are believed to be the “sparks” of AGI (Bubeck et al., 2023). Meanwhile, there is increasing concern about the potential safety risks of LLM agents when they surpass human capacity (Morris et al., 2023; Feng et al., 2024). To achieve safety and harmony in a future society where humans and AI agents with superhuman intelligence live together (Tsvetkova et al., 2024), we need to ensure that AI agents will cooperate, assist and benefit rather than deceive, manipulate or harm humans. Therefore, a better understanding of LLM agent trust behavior can help to maximize their benefit and minimize potential risks to human society.

7 Conclusion

In this paper, we discover LLM agent trust behavior under the framework of Trust Games, and behavioral alignment between LLM agents and humans regarding trust behavior, which is particularly high for GPT-4. This suggests the feasibility of simulating human trust behavior with LLM agents and paves the way for simulating human interactions and social institutions where trust is critical. We further investigate the intrinsic properties of agent trust under multiple scenarios and discuss broader implications, especially for social science and role-playing services. Our study offers deep insights into the behaviors of LLM agents and the fundamental analogy between LLMs and humans. It further opens doors to future research on the alignment between LLMs and humans beyond values.

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